

Exam 3 Practice

Practice Exam 3 – CE 2200, Dr. Koen

Section 1 – Internal forces at a point and equations.

1.1 Determine the normal force in the beam at point B.

$$N_B = 266.6 \text{ lb (+)}$$

1.2 Determine the shear force in the beam at point B.

$$V = 0$$

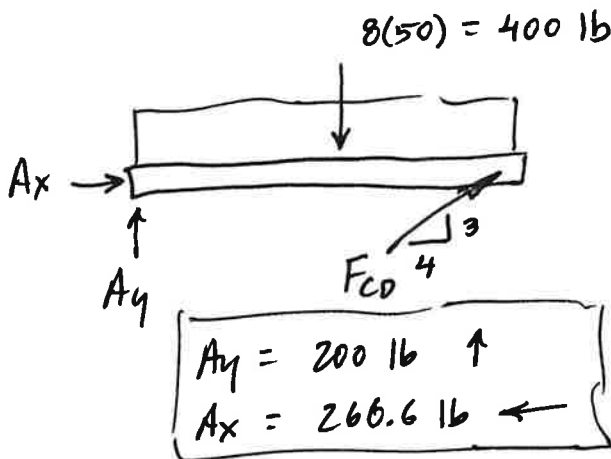
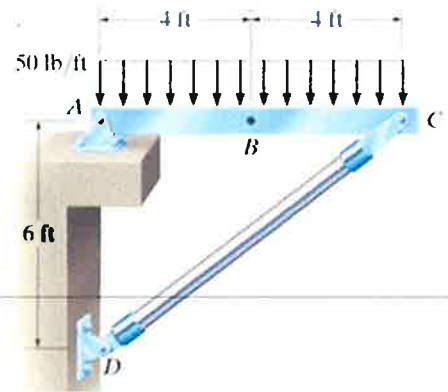
1.3 Determine the bending moment in the beam at point B.

$$M_B = +400 \text{ lb-ft}$$

1.4 What is the shear force at any point along member DC?

Zero

DC is a two-force member; only has axial force.

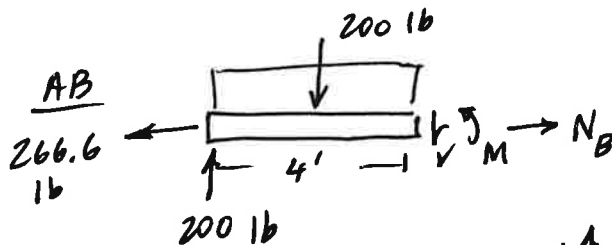


$$+\circlearrowleft \sum M_A = 0; 400(4) - \frac{3}{5} F_{CD}(8) = 0$$

$$F_{CD} = \frac{1600}{.6(8)} = 333.3 \text{ lb}$$

$=$

$\frac{3}{5} 333.3 = 200 \text{ lb}$
 266.6 lb



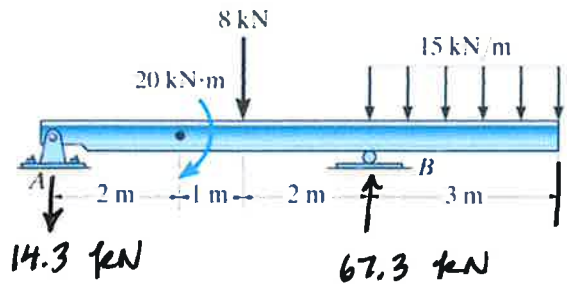
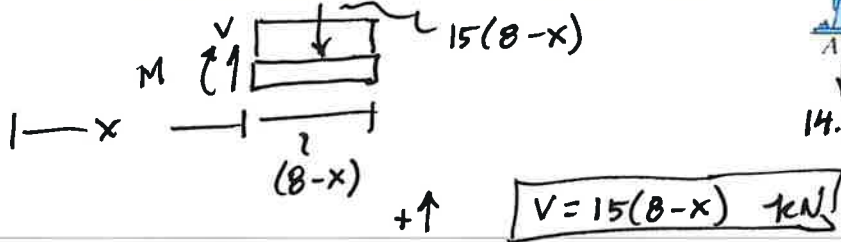
$$+\uparrow 200 - 200 - V = 0; \quad V = 0$$

$$+\circlearrowleft \sum M_B = 0; M + 200(2) - 200(4) = 0$$

$$M = +400 \text{ lb-ft}$$

1.5 Write the shear equation for the segment between $5 < x < 8$ (origin of x-axis is at point A).

Support Reactions $A_y = 14.3 \text{ kN} \downarrow$ and $B_y = 67.3 \text{ kN} \uparrow$



1.6 Write the moment equation for the segment between $5 < x < 8$ (origin of x-axis is at point A)

Support Reactions $A_y = 14.3 \text{ kN} \downarrow$ $B_y = 67.3 \text{ kN} \uparrow$

$$+\circlearrowleft \sum M_{\text{cut}} = 0; \quad M = -15(8-x)\left(\frac{8-x}{2}\right)$$

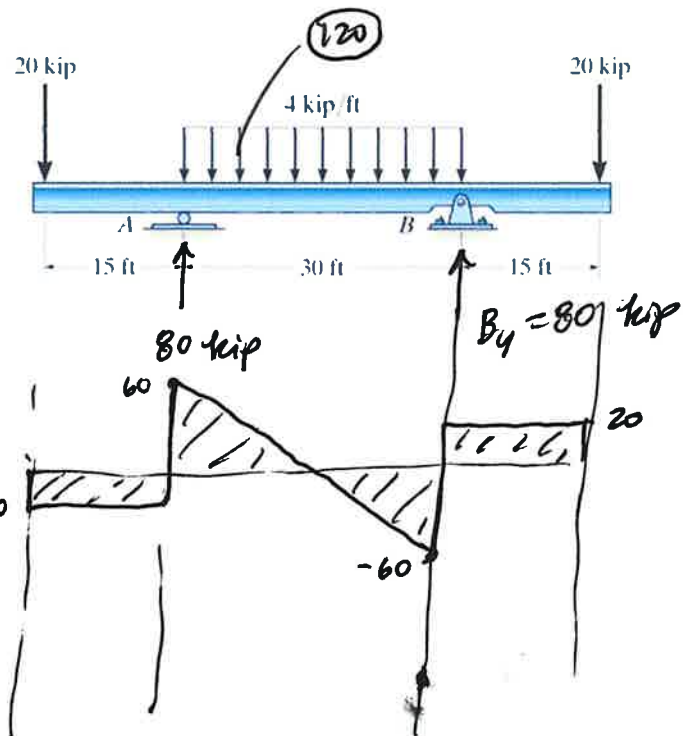
$$M = -7.5(8-x)^2$$

Section 2 – Graphing VM Diagrams with 6 rules

Support reactions.....

$A_y = B_y = 80 \text{ kip} \uparrow$

The x axis has its origin at the left side of the beam



2.1 What is the numerical value of change (magnitude) of the change in shear for the section $15 < x < 45 \text{ ft}$ and what is the line type for the connection?

-120 kip



2.2 What happens to the shear diagram at $x = 45 \text{ ft}$?

Jumps $\uparrow 80 \text{ kip}$

Support reactions.....

$$A_y = 14.3 \text{ kN} \downarrow \quad B_y = 67.3 \text{ kN} \uparrow$$

The x axis has its origin at the left side of the beam

2.3 What is the numerical value of change (magnitude) of the change in the moment diagram for the section $5 < x < 8 \text{ m}$ and what is the line type for the connection?

$$+ 67.5$$

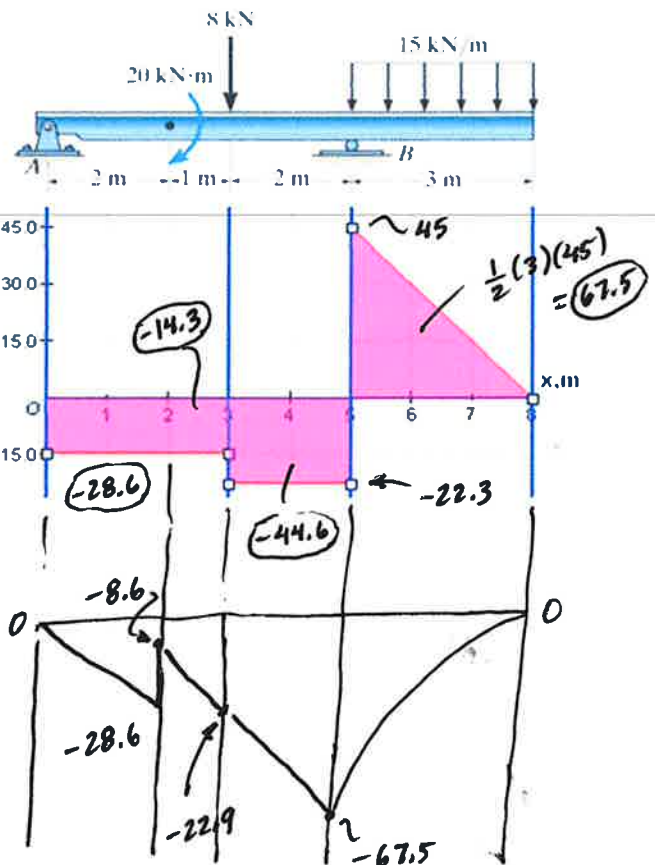
kN-m

2.4 What happens to the moment diagram at $x=2$?

Jumps $20 \text{ kN-m} \uparrow$

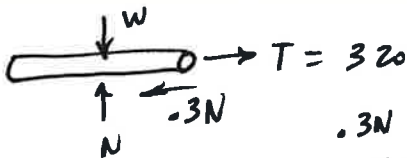
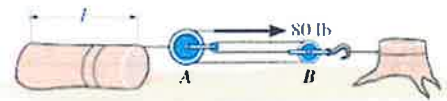
2.5 What happens to the moment diagram at $x=8$?

$$M=0$$



Section 3 – Friction

3.1 (5 pt) The log has a coefficient of static friction of $\mu = 0.3$ with the ground and a weight of 40 lb/ft . If a man can pull on the rope with a maximum force of 80 lb , determine the greatest length " L " of the log he can bring to the point of impending motion.



$$0.3N = 320$$

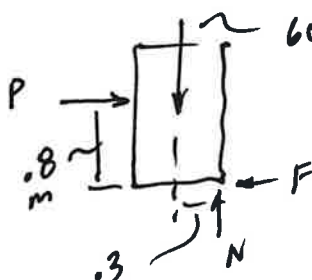
$$N = 1066.7 \text{ lb} = W = L \cdot 40$$

$$L = 26.7 \text{ ft}$$



$$4 \times 80 = 320$$

3.2 (4 pt) What value of P would cause tipping of the uniform 60-kg crate?

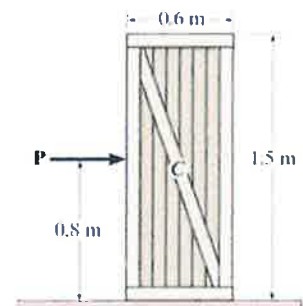


$$60(9.81) = 588.6 \text{ N}$$

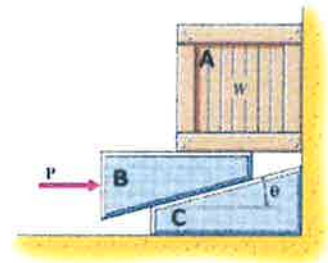
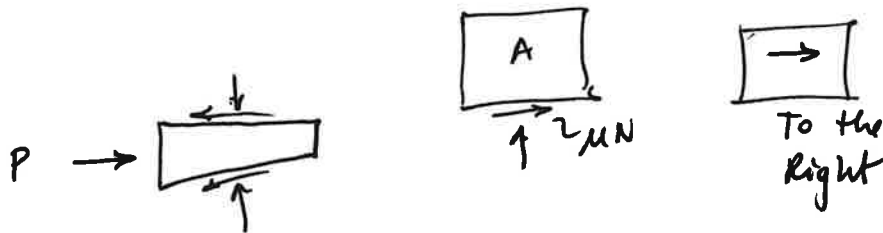
$$+\uparrow \sum M_N = 0;$$

$$P(0.8) - 588.6(0.3) = 0$$

$$P = 220.7 \text{ N}$$



3.3 (2 pt) For Block A, determine the direction of the friction force acting on the bottom of Block A as a result of the impending motion of Wedge B



When calculating the minimum force P required to prevent motion of the wheel....

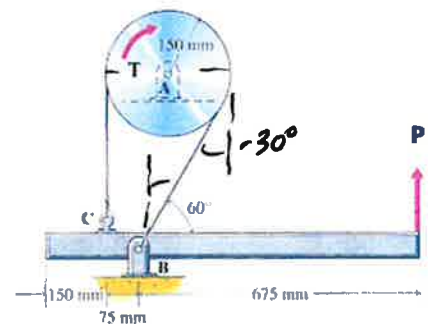
3.4 (4 pt) What is the angle beta to be used in the equation for belt friction when working with this system?

$$\theta = (180 + 30) \cdot \frac{\pi}{180}$$

3.5 (4 pt) Which tension would be T_2

$$\theta = 3.665 \text{ rad}$$

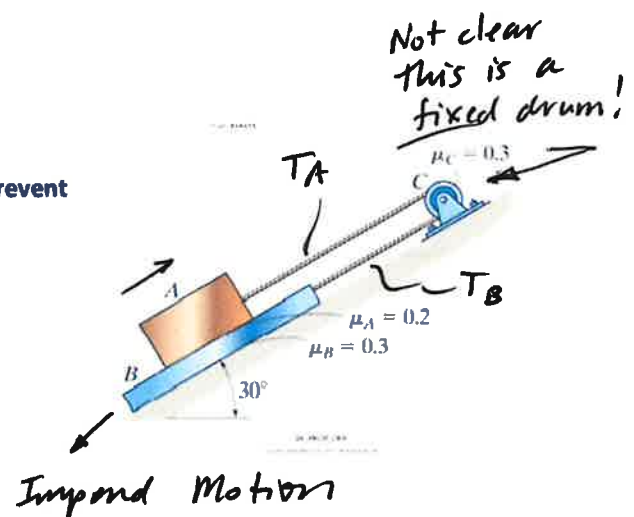
$$T_C = T_2$$



3.6 (6 pt) When determining the smallest mass of block A that would prevent motion, which tension would be T_2 and what would Beta equal?

$$T_B = T_2$$

$$\beta = \pi$$



Section 4 – Centroids and theorems of P&G

10 (pts) When finding the centroid by single integration with using a vertical element, which are the correct expressions?

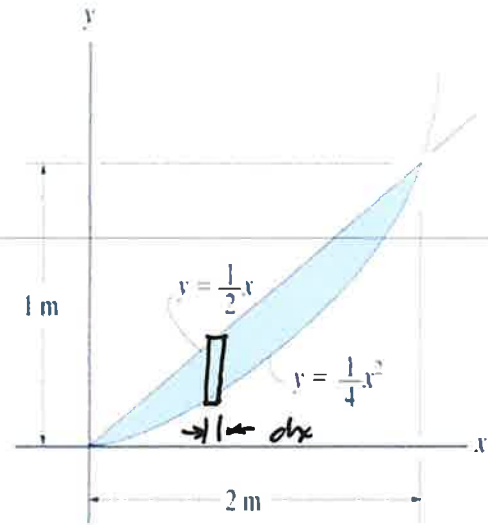
4.1 $\tilde{x} = x$

4.2 $\tilde{y} = \left(\frac{1}{2}x + \frac{1}{4}x^2 \right) \cdot \frac{1}{2}$

4.3 $dA = \left(\frac{1}{2}x - \frac{1}{4}x^2 \right) dx$

4.4 What are the limits for integration?

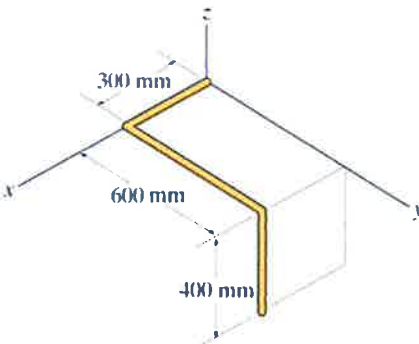
$$0 \leq x \leq 2$$



4.5 (5 pt) Determine the location of the centroid $\bar{y}$ -bar for the bent wire shape.

$$\bar{y} = \frac{\sum \tilde{y} L_i}{\sum L_i} = \frac{0 + .3(.6) + (.6)(.4)}{1.3 \text{ m}}$$

$$\bar{y} = 0.323 \text{ m} = 323 \text{ mm}$$



4.6 (10 pt) Determine the outside (not including the bottom) surface area and volume of the solid formed by a full rotation of the shaded area.

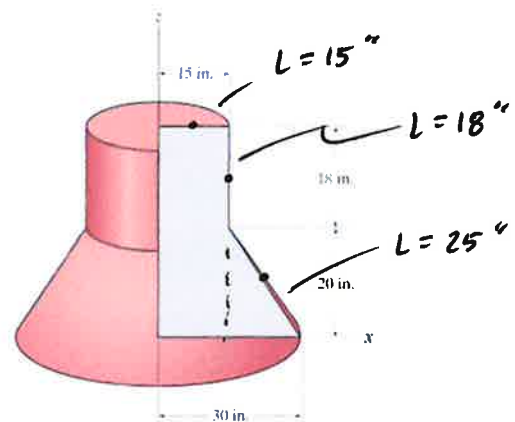
$$\theta = 2\pi$$

$$A = \sum \theta \bar{r} L = 2\pi [7.5 \cdot 15 + 15 \cdot 18 + 22.5(25)]$$

$$A = 5937.6 \text{ in}^2$$

$$V = 2\pi [7.5(570) + (15+5)(150)]$$

$$V = 45.71 \cdot 10^3 \text{ in}^3$$



$$A = (15)(38) = 570$$



$$A = \frac{1}{2}(15)(20) = 150$$

Answers 1.1 + 267 lb 1.2 0 lb 1.3 + 400 lb-ft 1.4 0 lb	3.1 26.7 ft 3.2 221 N 3.3 to the right 3.4 3.665 rad 3.5 cable connected at C
1.5 {120 - 15x} kN 1.6 {-480 + 120x - 7.5x²} kN-m 2.1 120 kip 2.2 jumps up 80 kip 2.3 67.5 kN-m 2.4 jumps 20 kN-m 2.5 no change	3.6 cable connected to block B pi radians 4.1 x 4.2 (.5x + .25x²)/2 4.3 (.5x - .25x²) dx 4.4 0 to 2 4.5 323 mm 4.6 surface area = 5938 in² vol = 45,710 in³